

Lab Report Guidelines

LAB 1: Exploratory Data Analysis on YouTube Data

Please complete Lab 1: Exploratory Data Analysis on YouTube Data following the provided manual. You are required to prepare and submit a lab report (PDF format) along with your Python files (.py or .ipynb).

Lab Report Guidelines

Your lab report should include the following sections:

1. Cover Page

2. Introduction

Briefly explain the goal of this lab and what you aim to explore (YouTube captions vs. comments analysis).

3. Tools and Setup

Mention the Python version, libraries used (pandas, matplotlib, webvtt-py, etc.), and the yt-dlp installation process.

4. Data Collection

- Mention the YouTube channel and video selected (must follow the criteria from the manual).
- Explain how you downloaded captions (.vtt) and comments (comments.txt).

5. Data Processing

Include code snippets for:

- Loading comments and captions
- Cleaning and preprocessing steps

6. Experiments and Results

Include the outputs (plots, statistics, or printed values) for each of the following:

- Histogram of caption vs. comment lengths
- Vocabulary diversity (Type-Token Ratio)

- Top-N word frequency (after stopwords removal)
- Any additional plots (caption/comment trends, etc.)

7. Analysis and Discussion

Interpret the results — compare caption and comment characteristics, and highlight key observations.

If you faced any issues while doing the experiments, mention them in the Discussion section of your report.

8. Conclusion

Summarize your findings and insights from this exploratory data analysis.

Submission Requirements:

Upload the following files:

1. Lab Report (PDF format)
2. Python code file (.py or .ipynb)
3. Captions and comments files used (captions.vtt, comments.txt)

Please make sure to submit your work before the deadline.